Data Center Resource Consumption

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Data Centers are Real, Physical Things

- The 'Cloud' isn't abstract. Digital actions are physical processes.
- Digital computing requires (among other things):
  - Energy to run
  - Water to cool*
  - Space and support infrastructure
- The energy grid used to provide power also affects water/GHG
- Global AI adoption is making these effects non-negligible

Computing and Energy

- Digital electronics rely on electrical energy
  - Change and maintain voltage states
  - AI primarily using GPUs
- Inefficiencies lead to wasted energy (heat)
- DC energy consumption could be ~12% of US total by 2028

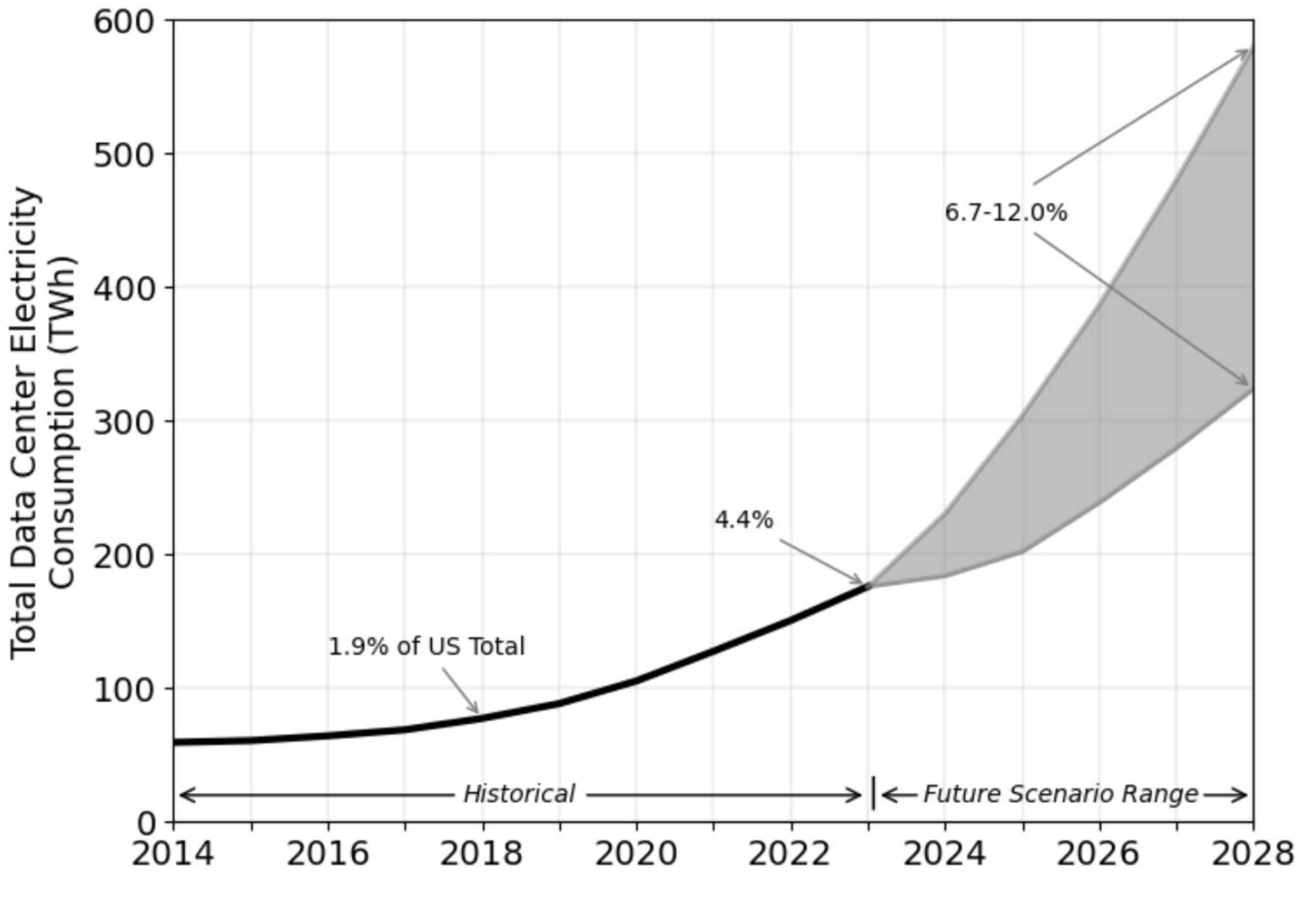


Figure ES-1. Total U.S. data center electricity use from 2014 through 2028.

Source: 2024 United States Data CenterEnergy Usage Report

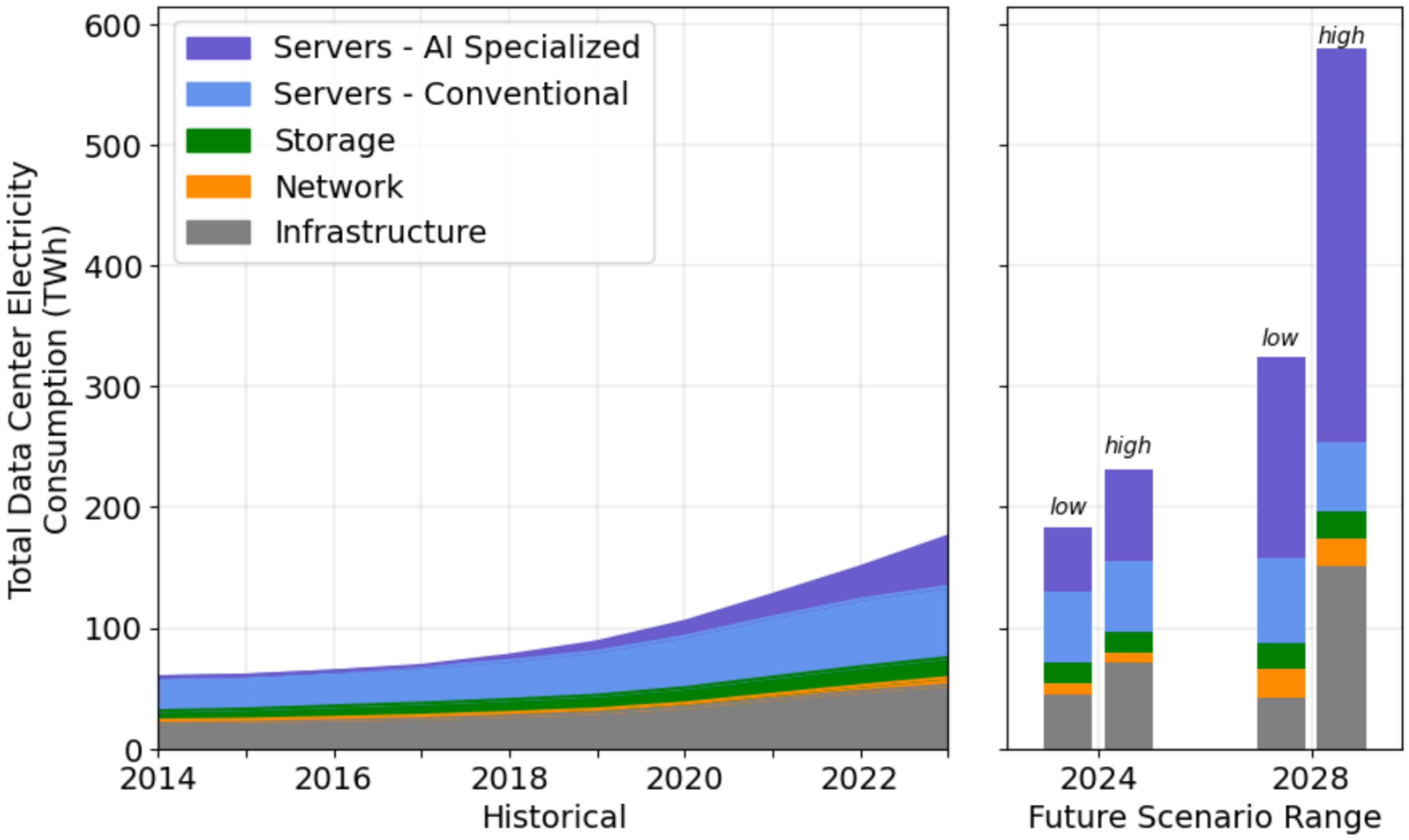


Figure 5.6. Total data center electricity use from 2014 through 2028 by equipment type.

Source: 2024 United States Data CenterEnergy Usage Report

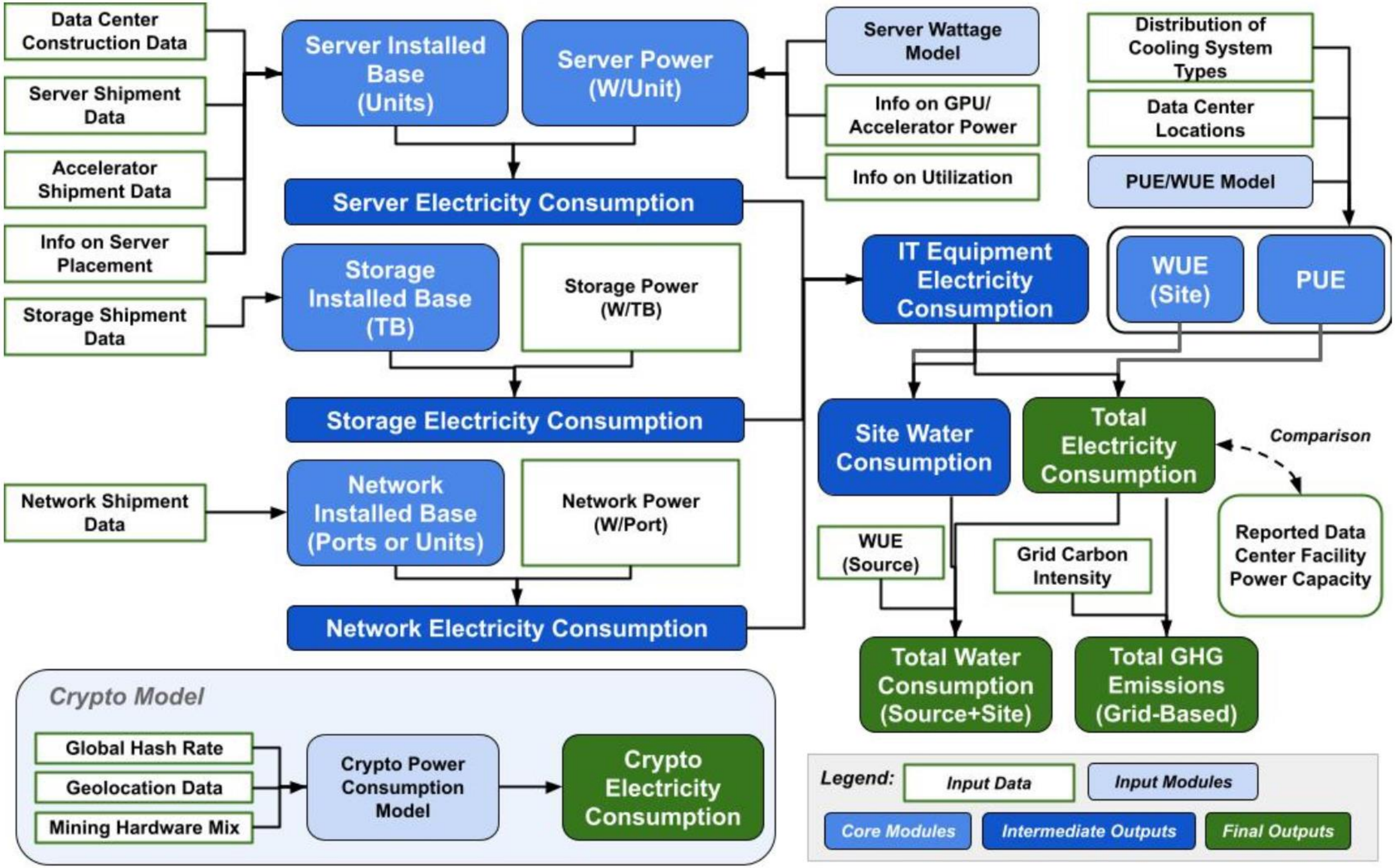


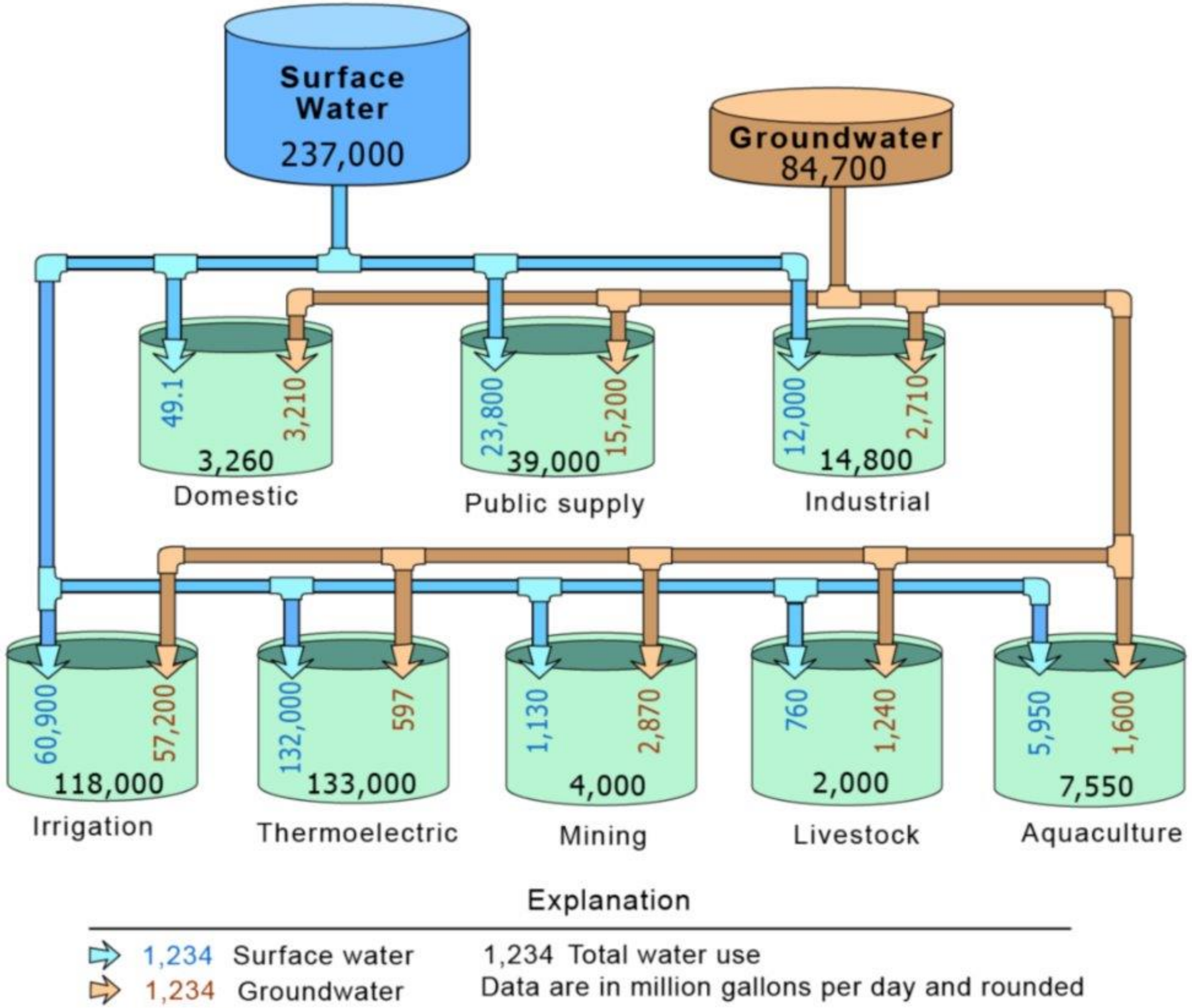
Figure 2.1. Flow chart for the data center electricity model used in this study.

Source: 2024 United States Data CenterEnergy Usage Report

Importance of the Energy Grid

- DCs directly use water, but also indirectly
  - Primarily through the energy grid that powers them
- Thermoelectric power requires cooling (and water)
  - In the US, it's the largest consumer of water! (USGS – see next page)
- Carbon footprint is heavily dependent on energy generation

USGS 2015 Water Report Visual Summary



Water and GHG Impacts of the Energy Market

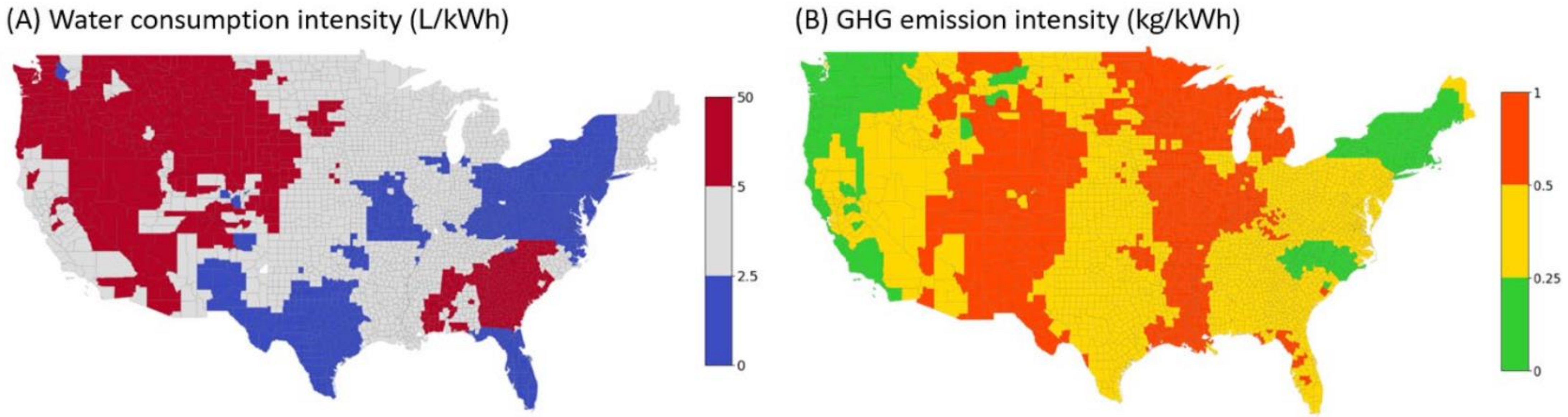
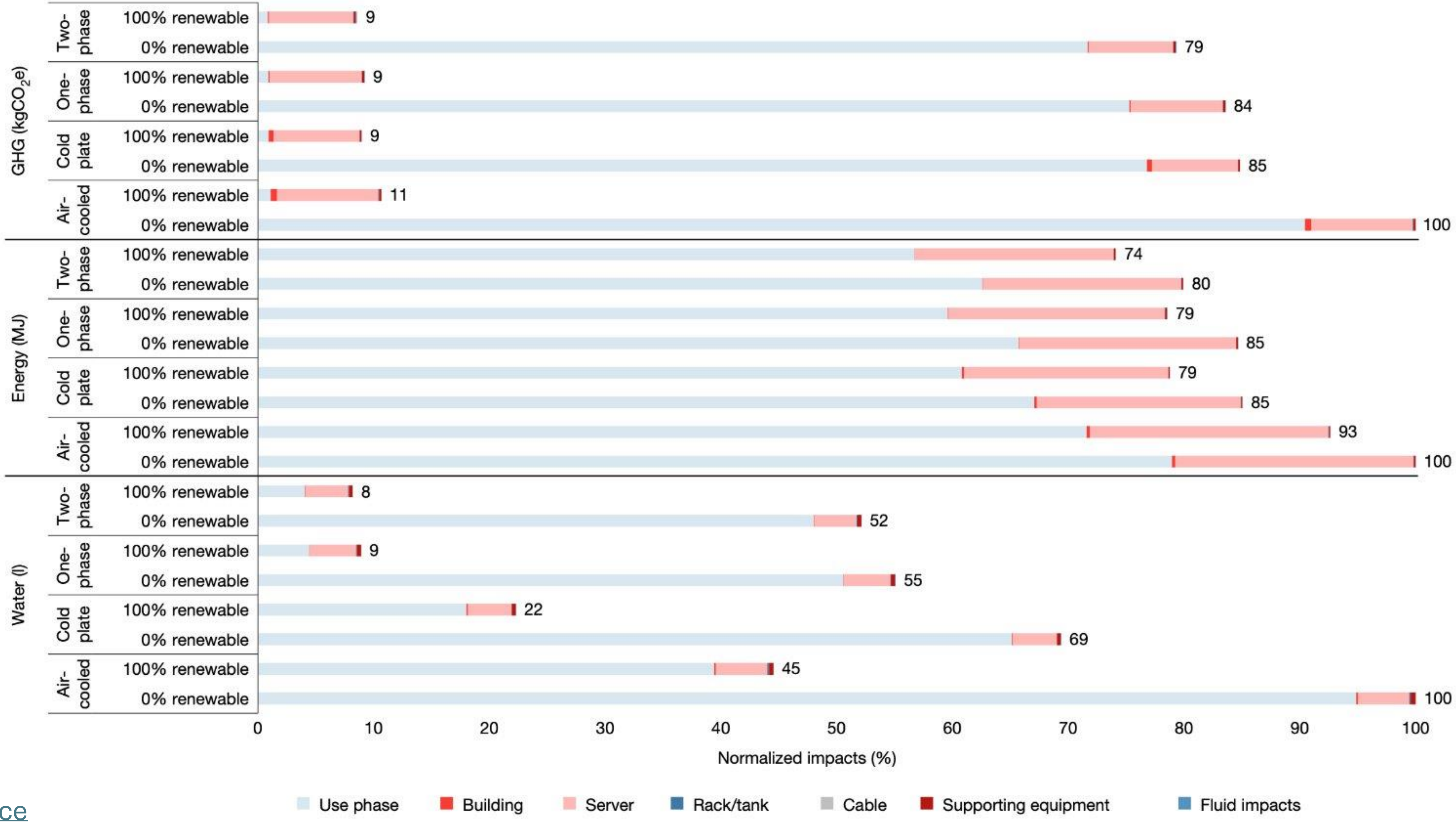


Figure 5.10. Water consumption and GHG emission intensity factors of electricity use by county.

[Source: 2024 United States Data CenterEnergy Usage Report]



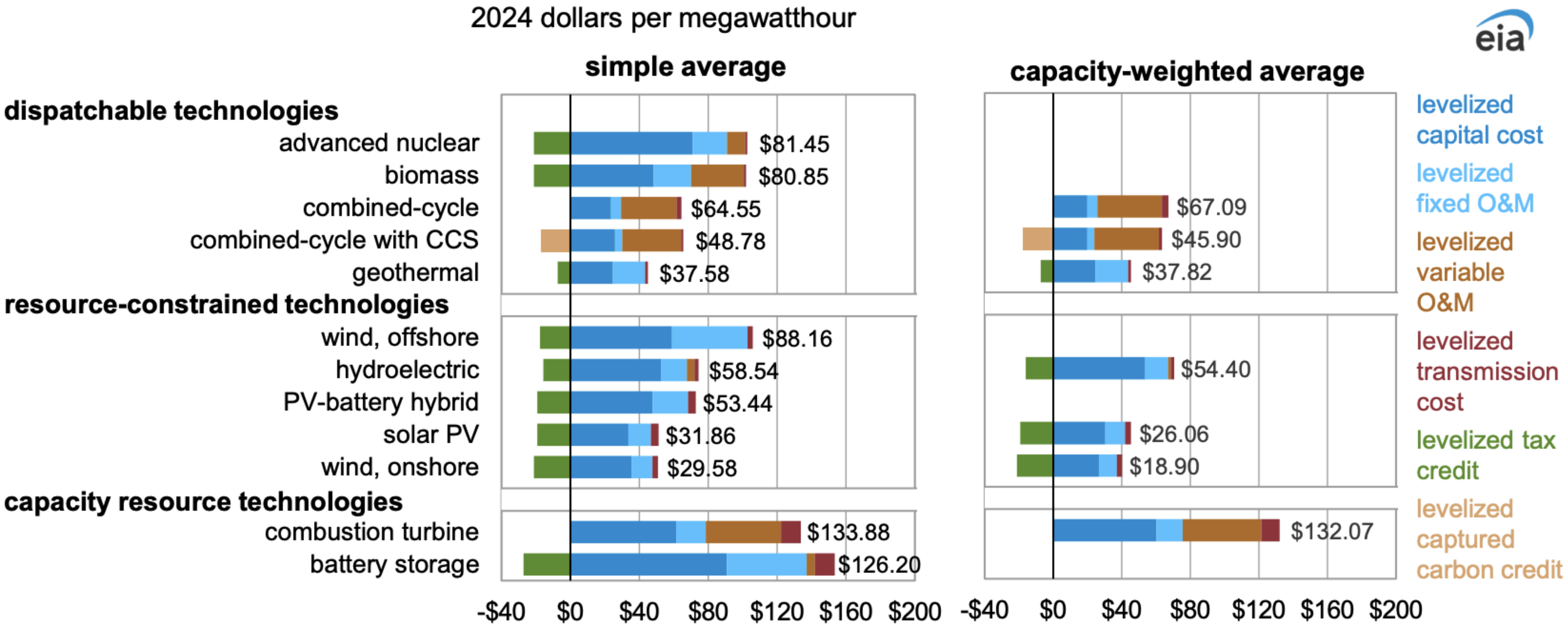
GHG (kgCO2e)
Energy (MJ)
Water (l)
Two-phase
One-phase
Cold plate
Air-cooled
100% renewable
0% renewable
9
79
9
84
9
85
11
100
74
80
79
85
79
85
93
100
8
52
9
55
22
69
45
100
0
10
20
30
40
50
60
70
80
90
100
Normalized impacts (%)
Use phase
Building
Server
Rack/tank
Cable
Supporting equipment
Fluid impacts

Source

Cost?

- Monetary costs of running/paying for AI services matter
- Energy prices have a large impact on AI cost
    - Ignore market capturing forces currently dominating (maybe)
- Decreasing AI inference costs mean minimizing energy, water, and carbon costs

Estimated levelized cost of electricity (LCOE) and levelized cost of storage (LCOS) for new resources entering service in 2030



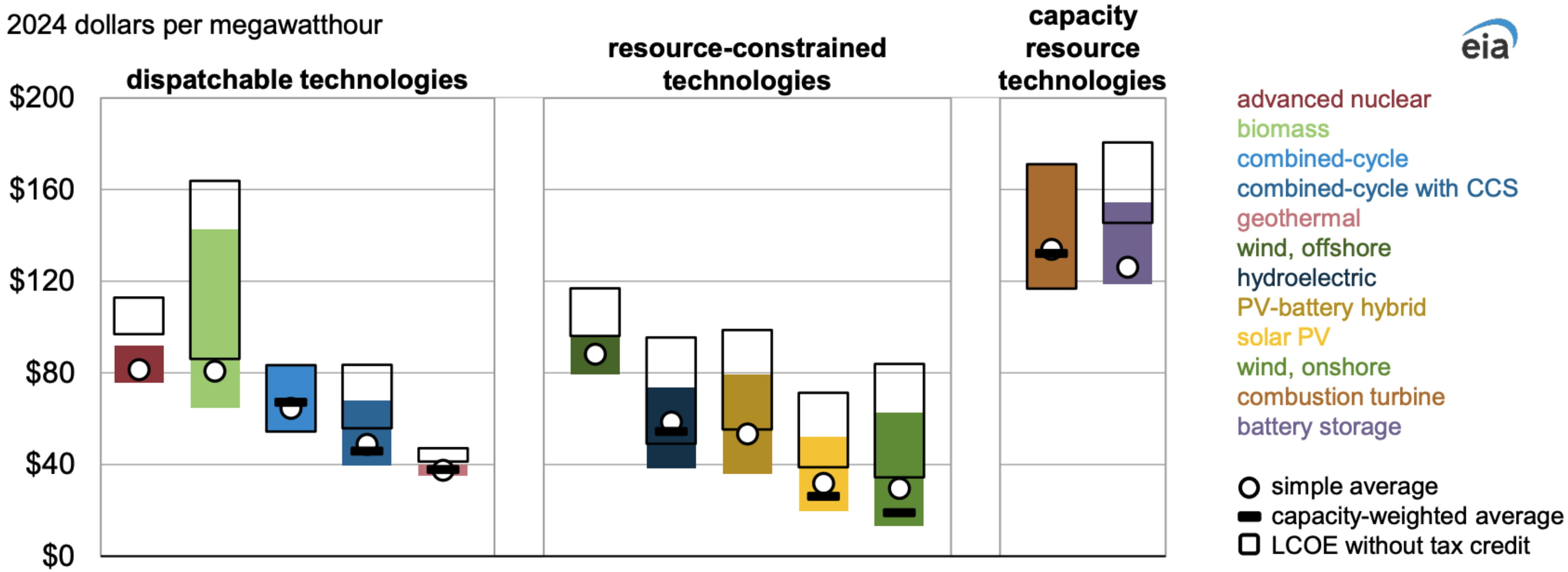
Data source: U.S. Energy Information Administration, Annual Energy Outlook 2025

Note: Technologies with no additions in 2030 do not have a capacity-weighted average. The stated LCOE values include the levelized tax credit component for eligible technologies. CCS=carbon capture & sequestration, PV=photovoltaic, O&M=operations and maintenance

Source: U.S. Energy Information Administration

Solar PV LCOE is lower than natural gas combined-cycle LCOE on average and, in most regions, even without the tax credit

Regional variation in levelized cost of electricity (LCOE) and levelized cost of storage (LCOS) for new resources entering service in 2030 by technology, AEO2025 Reference case



Data source: U.S. Energy Information Administration, Annual Energy Outlook 2025

Note: Technologies with no additions in 2030 do not have a capacity-weighted average. CCS=carbon capture & sequestration, PV=photovoltaic

Source: U.S. Energy Information Administration

Renewables and Demand Flexibility

- If these resources are not always available, flexibility is needed
  - Lack of power due to variable generation
  - Lack of water due to drought
  - Carbon capture limitations
- Computing demand can be easily reduced/altered
  - Emerald AI – a start-up in this space

What is the limit of resource consumption for computing use?